

Ryder Sitcawich

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EDUCATION

University of Pennsylvania

May 2026

Bachelor of Science in Engineering in Computer Science and Philosophy, Minor in Math

GPA: 3.97

- **Relevant Coursework:** Applied Machine Learning, Distributed Systems, Artificial Intelligence, Operating Systems, Advanced Databases, PL Theory, Computer Architecture, Scalable Cloud Computing, Security

EXPERIENCE

Meta

May – Aug 2025

Software Engineer Intern

New York, New York

- Developed and launched AI Background Themes on Messenger Web using React, GraphQL, and LightSpeed DB, resulting in **10,000+** daily themes generated and **37%** increase in MetaAI prompter DAU
- Owned launch end-to-end: designed A/B experiments, analyzed engagement metrics, coordinated cross-functional QA and dogfooding, and drove go/no-go decision resulting in launch approval
- Engineered a media send architecture for Edit Images with AI on Messenger Web, bypassing the message composer via LightSpeed DB Stored Procedures with efficient image processing, optimistic local sends for responsive UX, and server-confirmed delivery

UPenn CIS Department

Jan. 2024 – Present

Head Teaching Assistant, [CIS 1200](#) (Programming Languages and Techniques)

Philadelphia, PA

- Lead **40+** TA team supporting **400** students per semester; coordinate grading, office hours, and recitations
- Design and conduct weekly new TA training, mentoring new staff on teaching and debugging best practices

PROJECTS

Datacenter Cooling Optimization (Senior Capstone) | *Java, JavaFX, PyVISA*

Sep. 2025 – May 2026

- Designed and built a datacenter simulation integrating thermosensitive hydrogel passive cooling with intelligent computational sprinting
- Developed Bellman-equation oracle to optimize computational throughput given rack-level power constraints and chip-level thermal constraints
- Implemented hardware-in-the-loop testing via PyVISA/SCPI control of a Tektronix power supply to validate thermal models against physical hydrogel behavior

[PennCloud](#) | *C++, gRPC, HTML, CSS*

Mar. – May 2025

- Engineered a full-stack cloud platform inspired by Google Drive and Gmail, built on a Bigtable-style architecture with tablet partitioning, primary replication, and heartbeat fault-detection across distributed backend nodes
- Implemented a distributed key-value store with write-ahead logging and periodic checkpointing to ensure durability, low-latency access, and crash recovery under concurrent request loads
- Integrated frontend services (webmail, drive, account) with round-robin load balancing on both frontend and backend tiers, cookie-based session management, and a real-time admin console for node health monitoring

PennOS | *C*

Mar. – May 2024

- Implemented a FAT16-based file system with standard POSIX-style operations and a priority scheduler with three-level queuing and fixed quanta for fair process execution (2.5k lines of code)
- Integrated kernel and file system layers via shell built-ins and system calls using process-level file descriptor tables interfacing with a global descriptor table

TECHNICAL SKILLS

Languages: Java | C | C++ | OCaml | TypeScript | JavaScript | SQL | HTML/CSS | Python | HCL | Rocq (Coq)

Technologies: React | AWS | gRPC | Terraform | Docker | Node | ChromaDB | Next | Firebase | Django | PostgreSQL | Pandas | NumPy | TensorFlow

Interests/Misc.: Penn Labs | Travelers SWE | Sigma Alpha Mu Fraternity (Academics Chair) | Soccer | Cooking | Poker | Weightlifting